

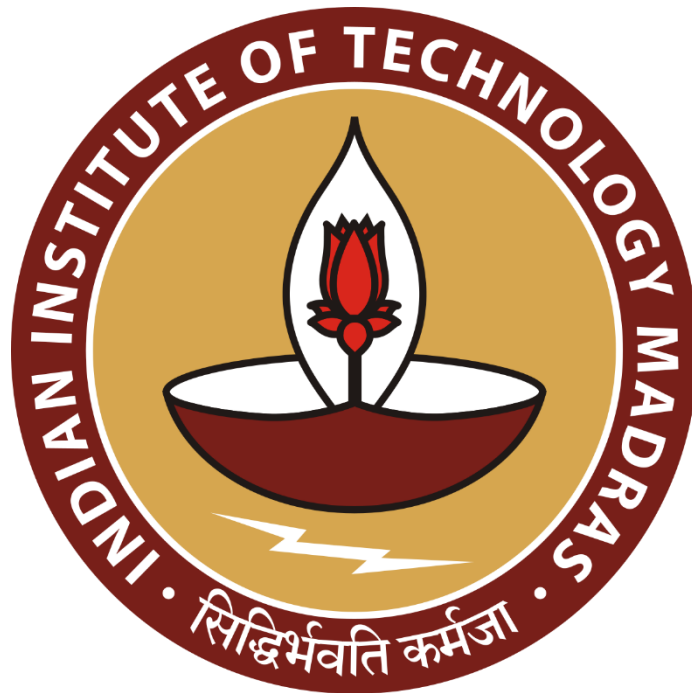
# Daily sales analysis of a café to increase efficiency and profits

## A End-Term report for the BDM capstone Project

Submitted by

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# 1 Executive Summary

This project analyses Coffee Beans, a small café located in Anand Vihar, Delhi. The café owner maintains handwritten records of daily sales and expenses which had never been formally analysed. When asked, the owner didn't know how well his business was actually doing and what he could do to make it better. This report tries to help him out with that. I found out 2 core problems: income unpredictability and expense driven profit instability.

A total of 150 calendar days of register data were digitised, of which 133 valid operating days were identified after excluding closure days. Across these days, average daily sales were ₹1,490.73 with a standard deviation of ₹955.37. Average daily expense was ₹1451.14, and average daily net profit was ₹39.59, indicating that the café operated at a marginal net profit on average over the observed period. Of the 133 operating days, 108 (81.2%) were profitable while 25 (18.8%) recorded losses. March was the only month with a positive total net profit of ₹32,132, while April recorded the largest monthly loss of ₹-15,212.

To address Problem 1 (income unpredictability), a 7-day moving average was applied to daily sales to identify demand trends, break-even analysis was conducted to assess sustainability, and weekday-wise sales patterns were examined.

To address Problem 2 (expense-driven profit instability), expense-to-sales ratios and profit margin analysis were performed, the coefficient of variation (CV) was calculated for both sales and net profit, and loss magnitude analysis was conducted to identify extreme cost events. Sales showed a CV of 0.64 (standard deviation is 64% of the mean), while net profit showed a CV of 91.38 (standard deviation is 91 times the mean), reflecting extreme instability at the earnings level.

The findings show that the café's financial instability is not driven by weak demand but by irregular and poorly timed expense spikes, particularly bulk salary and rent payments on single days. Recommendations focus on restructuring fixed cost timing, aligning procurement with high-demand windows, and implementing adaptive daily expense controls.

## 2 Detailed Explanation of Analysis Process/Method

This section explains how the café's handwritten records were transformed into structured financial analysis. The objective was to use relevant and logical analytical methods that directly address the café's issue of income unpredictability and unstable profit.

### 2.1 Data Preparation and Operating Day Definition

The primary source of data was the café's handwritten daily register. I digitized 150 calendar days of entries into Excel. However, not every recorded date represented actual business activity. Some days had both Sales and Expense recorded as zero or were left blank.

To ensure realistic analysis, I defined an **Operating Day** as any day where at least one of the following was recorded:

- Sales greater than zero
- Expense greater than zero

Days where both values were zero or missing were excluded from statistical calculations. This prevents distortion in averages and volatility measures caused by closure days.

After applying this rule, 133 operating days remained for further analysis.

Profit for each operating day was calculated using:

Net Profit = Sales – Total Expense

The “Total Expense” is a composite figure that includes the cost of raw materials, worker salary, rent, delivery costs, and all other operational expenditure recorded for the day.

This ensured consistency across all observations.

### 2.2 Descriptive Statistical Analysis

To understand the overall financial behaviour of the café, I calculated descriptive statistics for Sales, Expense and Net Profit. These included mean, median, standard deviation, minimum and maximum values.

The mean provides the overall average performance, while the median represents a typical day without being heavily influenced by extreme expense spikes. Standard deviation measures how much daily performance fluctuates around the average.

| Descriptive statistics (operating days only) |         |          |            |
|----------------------------------------------|---------|----------|------------|
|                                              | Sales   | Expense  | Net Profit |
| count                                        | 133.00  | 133.00   | 133.00     |
| mean                                         | 1490.73 | 1451.14  | 39.59      |
| std                                          | 955.37  | 3648.19  | 3617.99    |
| min                                          | 150.00  | 0.00     | -23362.00  |
| 25%                                          | 910.00  | 141.00   | 190.00     |
| 50%                                          | 1290.00 | 370.00   | 690.00     |
| 75%                                          | 1820.00 | 916.00   | 1343.00    |
| max                                          | 6200.00 | 27267.00 | 4609.00    |

Figure 1: Descriptive statistics of Sales, Expense and Net Profit

This table helps establish a numerical foundation for the rest of the analysis. It shows that while a typical day may be profitable, variability is high, indicating instability in operations.

## 2.3 Time-Series Trend Analysis

Since the data is recorded daily, it forms a natural time series. To examine patterns over time, I plotted Sales, Expense and Net Profit against Date.

However, daily values fluctuate significantly. To reduce short-term noise and observe broader patterns, I calculated a 7-day moving average for Sales. This smoothing technique highlights weekly demand cycles without using complex forecasting models.

The moving average is appropriate because café operations often follow weekly footfall behaviour. It helps separate random fluctuations from consistent trends.

## 2.4 Expense Behavior and Ratio Analysis

Looking only at absolute expense values is not sufficient, since a high cost may still be reasonable on a high-sales day. Therefore, I calculated proportional measures:

- Expense-to-Sales Ratio = Expense / Sales
- Net Profit Margin = Net Profit / Sales

These ratios evaluate spending relative to revenue and help identify days where costs were disproportionately high.

| === Expense ratio & net Profit margin statistics === |               |                   |
|------------------------------------------------------|---------------|-------------------|
|                                                      | Expense_ratio | Net Profit_margin |
| count                                                | 133.0000      | 133.0000          |
| mean                                                 | 1.1284        | -0.1284           |
| std                                                  | 2.8279        | 2.8279            |
| min                                                  | 0.0000        | -18.6226          |
| 25%                                                  | 0.1003        | 0.2884            |
| 50%                                                  | 0.3107        | 0.6893            |
| 75%                                                  | 0.7116        | 0.8997            |
| max                                                  | 19.6226       | 1.0000            |

Figure 2: Expense ratio and Net Profit margin statistics

## 2.5 Volatility Measurement

To assess stability, I calculated standard deviation and coefficient of variation for both sales and profit.

The coefficient of variation measures variability relative to the average level and is calculated as:

$$CV = (\text{Standard Deviation} / \text{Mean})$$

A higher CV indicates greater instability. For example, a CV of 0.64 for sales means the standard deviation is 64% of the average daily sales. For net profit, the CV of 91.38 indicates that the standard deviation is over 91 times the average net profit. This extreme value occurs because the mean net profit is very small, making even moderate fluctuations appear disproportionately large.

This is important because even if the average profit appears reasonable, high variability makes planning and forecasting difficult.

Additionally, I calculated the percentage of profit-making days and loss-making days to measure how frequently the café experiences financial stress.

## 2.6 Break-even Assessment

To understand operational sustainability, I calculated the daily break-even sales level as:

$$\text{Break-even Sales} = \text{Average Daily Expense}$$

This represents the minimum revenue required to avoid losses under normal cost conditions.

I then identified the number and percentage of days where sales fell below this threshold. This provides a clearer measure of structural income risk.

## 2.7 Risk Clarification

To quantify instability further, I defined a High-Risk Day as:

- Net Profit less than zero
- OR
- Expense-to-Sales Ratio greater than 80%

This classification captures both realised losses and near-loss stress situations. Measuring the frequency of such days provides a clearer understanding of operational risk than relying solely on averages.

### 3 Results and Findings

This section presents the outcomes of the analysis performed on the 133 operating days. The findings are organized around the two core problems identified: income unpredictability and expense-driven profit instability.

#### 3.1 Problem 1: Income Unpredictability and Demand Instability

##### 3.1.1 Sales Behavior and Demand Stability

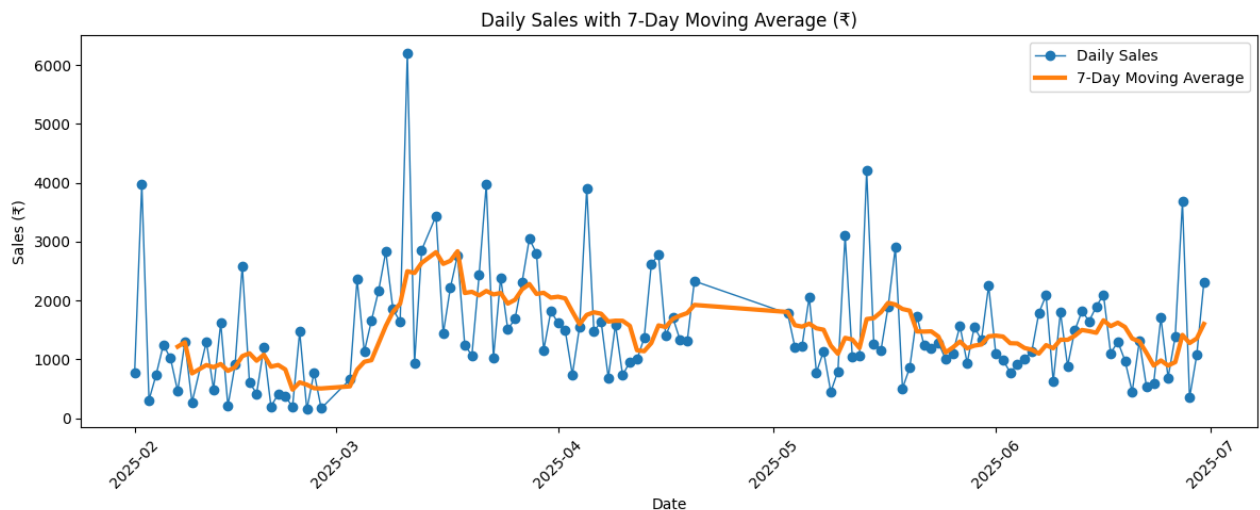


Figure 3: Daily Sales with 7-day moving average

As we can see in Figure 3, the daily sales line shows sharp fluctuations across the observed period. There are multiple high spikes as well as sudden drops, indicating that customer demand is inconsistent from one day to another.

When we observe the 7-day moving average, the pattern is smoother. Sales appear to rise during certain periods and fall during others, but there is no strong upward or downward trend. This suggested that demand exists, but it is irregular and not easily predictable on a daily basis.

The absence of a stable growth trend directly connects to the owner's difficulty in forecasting weekly or monthly income. Since daily sales fluctuate significantly, planning inventory based purely on recent days becomes risky.

This confirms the first problem statement: income unpredictability is structurally visible in the data.

### 3.1.2 Weekday Sales Performance

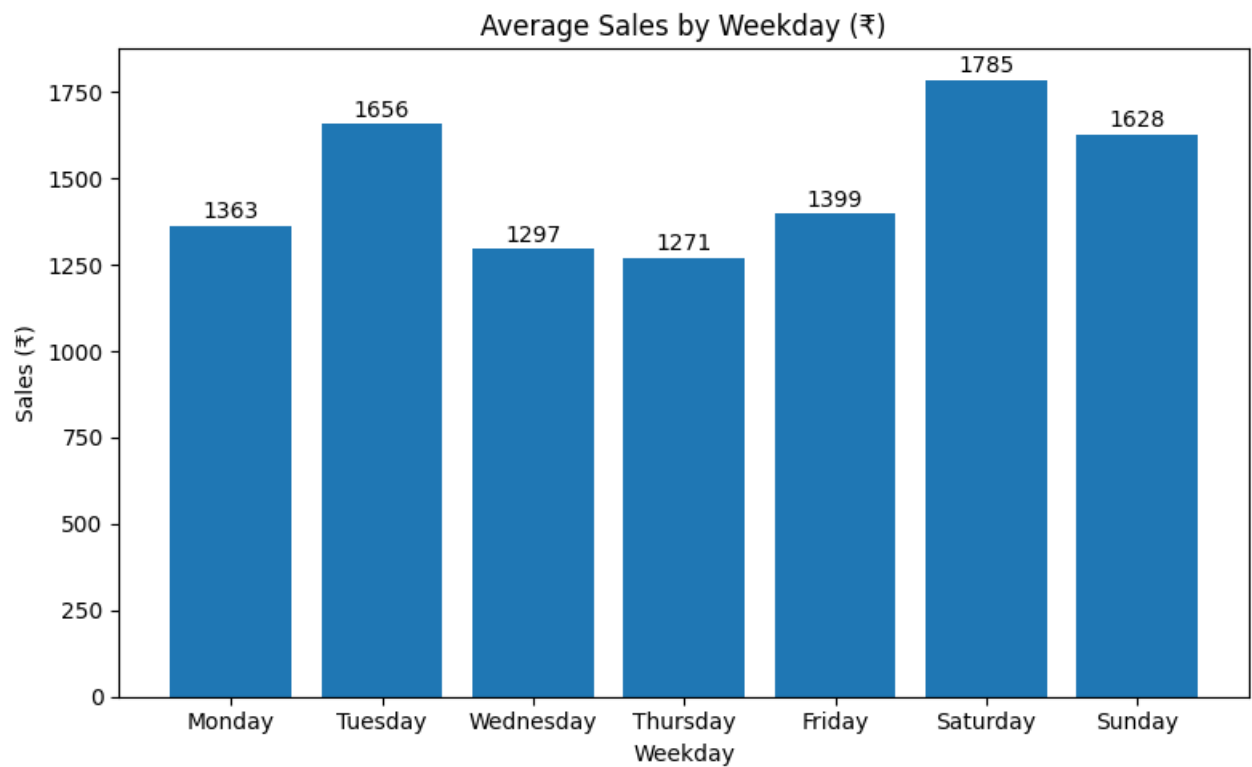


Figure 4: Average Sales by Weekday

As we can see in Figure 4, Saturday has the highest average sales, followed by Tuesday and Sunday. Thursday and Wednesday record comparatively lower average sales levels. This confirms that customer footfall is not evenly distributed across the week and tends to increase towards the weekend.

This confirms the problem statement that income patterns are uneven across the week, making forecasting difficult.

This suggests that inventory procurement and staffing decisions should be adjusted to reflect the uneven distribution of customer footfall across the week.

### 3.1.3 Break-even Performance Analysis

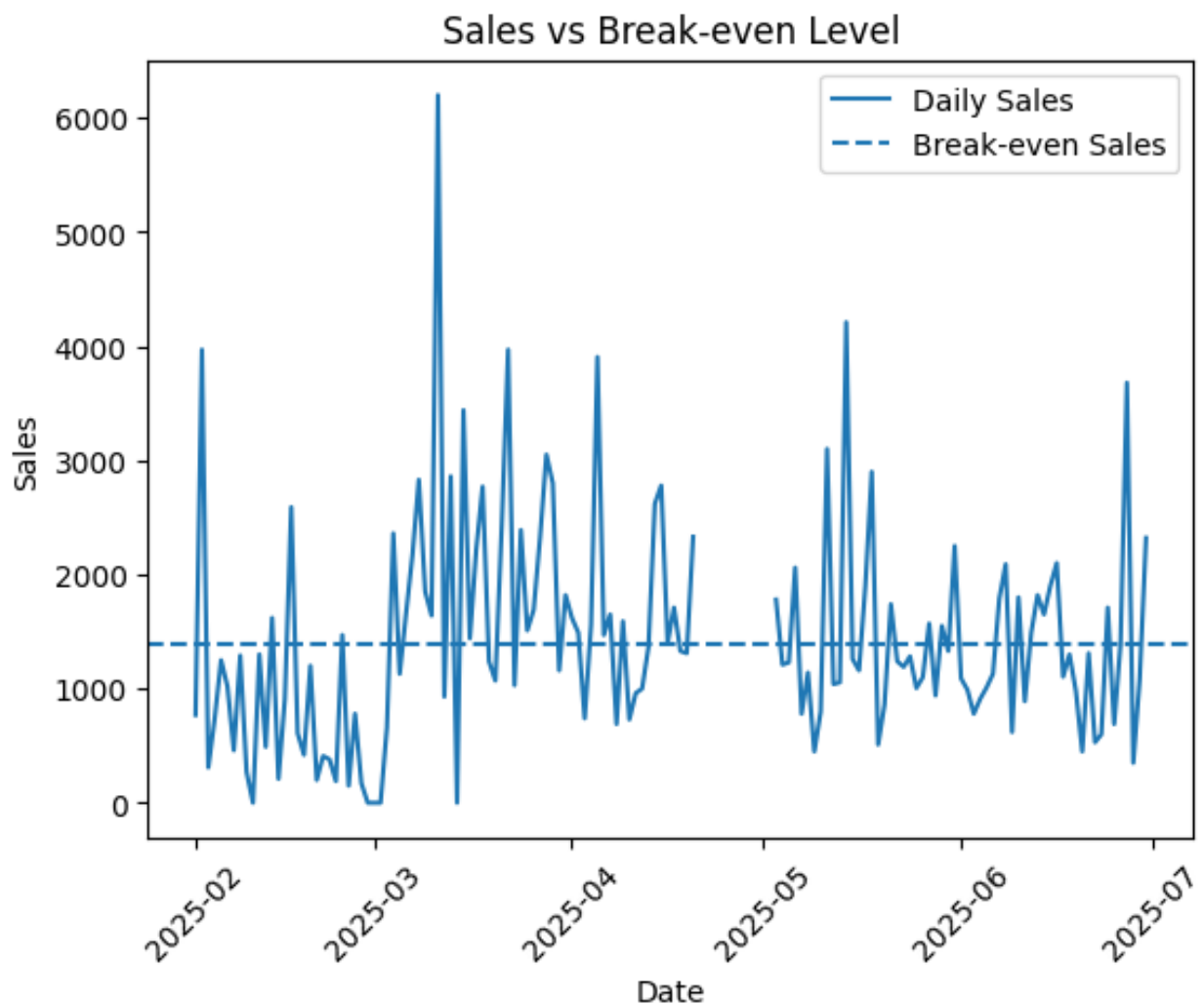


Figure 5: Sales vs Break-even Level

The break-even level represents the minimum sales required to cover average daily expenses.

Notably, the average daily sales of ₹1,490.73 exceeds the break-even level of ₹1,451.14 by only ₹39.59, confirming that the café operates with almost no margin of safety on a typical day.

As shown in Figure 5, the café falls below this threshold on a number of days. Even on days where sales exceed break-even, the margin above it is often small, leaving limited buffer for unexpected costs.

This indicates that the café operates close to its sustainability limit. Small increases in expense or minor drops in sales can cause losses.

## 3.2 Problem 2: Expense Behavior and Profit Instability

### 3.2.1 Relationship Between Sales and Expense



Figure 6: Expense vs Sales with loss days

The scatter plot in Figure 6 compares daily sales (x-axis) with daily expense (y-axis). Each blue point represents a normal operating day, while red points represent loss-making days.

Ideally, most observations should lie below an implied diagonal where expenses are lower than sales. However, the graph shows several days where expenses significantly exceed revenue. These red-highlighted points are concentrated at moderate sales levels but unusually high expense values.

This indicates that losses are not necessarily caused by low sales alone. In multiple instances, expenses spike disproportionately compared to revenue. These expense spikes appear to be the primary drivers of negative profit.

The clustering of red points at high expense levels suggests that instability is driven more by irregular cost events rather than absence of demand.

Seeing these red points sitting so high in the scatter plot, I went back to the data to find out the reason and figure out what was happening in particular on these days. I found that the 27,267 spike on 05-April was not particularly a bad day for the café with sales of ₹3905. The spike was caused

by rent and salary both being paid on the same day. Similarly, the spikes on 06-Feb, 05-May and 10-June were all caused by salary payout. The pattern became clear after this. The losses weren't unaccounted for or random. Losses are predictable and can be planned for in this case knowing that the fixed cost is going to hit.

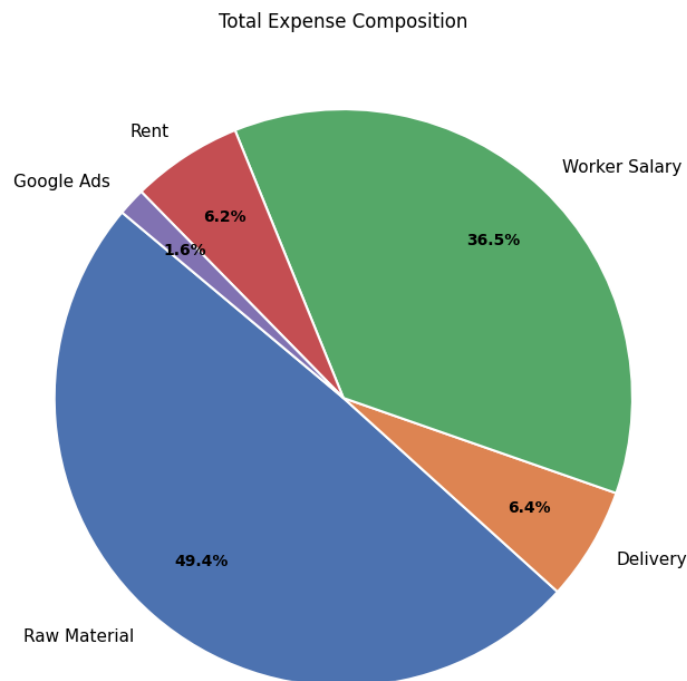


Figure 7: Expense Distribution across categories

As we see in figure 7, raw materials account for 49.4% of total expenses, representing the day-to-day operational cost of running the café. Worker salary is the second largest component at 36.5%, but unlike raw materials, salary payments are concentrated on a small number of days (10 out of 133 operating days), making them the primary driver of expense spikes. Delivery costs (6.4%) and rent (6.2%) are smaller but similarly irregular. Google Ads (1.6%) represents a one-time marketing expense. This composition confirms that on a typical operating day, expenses are manageable and proportional to sales. The instability arises specifically from the irregular timing of fixed cost payments, particularly salary and rent, which land as lump sums on single days rather than being distributed across the month.

### 3.2.2 Weekday Net Profit Performance

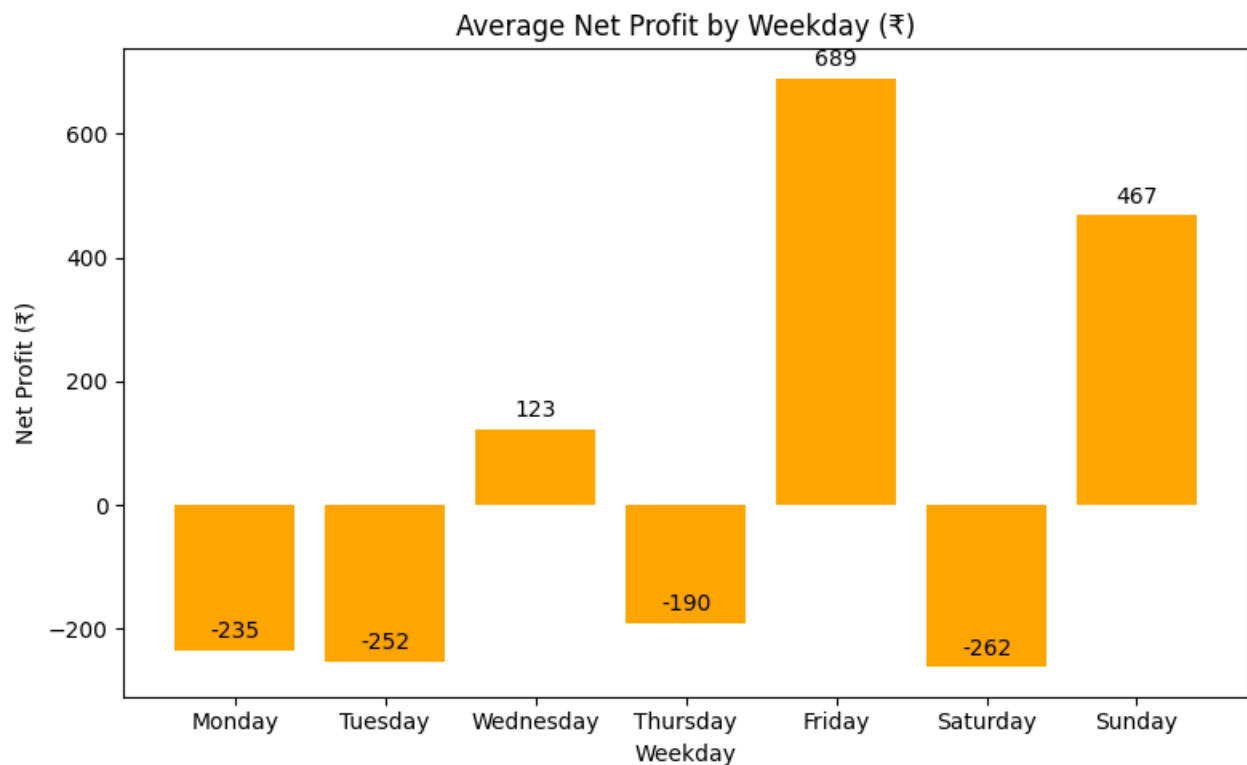


Figure 8: Average Net Profit by Weekday

Figure 8 tells us a different story compared to Figure 4. Despite strong sales on Tuesday and Saturday, both days show negative average profit. Monday and Thursday also record negative average profits. In contrast, Friday shows the highest average profit even though its sales are not the highest of the week.

This mismatch between sales and profit indicates that higher revenue does not automatically translate into better financial performance. The negative average profits on certain high-sales days suggest that expense spikes or inefficient cost allocation may be concentrated on those days.

The weekday analysis therefore strengthens the problem statement. First, income patterns are uneven across the week, making forecasting difficult. Second, spending behaviour appears inconsistent, leading to losses even on days with reasonable demand.

This suggests that cost planning and operational decisions may need to be adjusted based on weekday performance patterns.

### 3.2.3 Net Profit Distribution

| Net Profit distribution across operating days |                |      |         |
|-----------------------------------------------|----------------|------|---------|
|                                               | Category       | Days | Share_% |
| 0                                             | Net Profit > 0 | 108  | 81.2    |
| 1                                             | Net Profit < 0 | 25   | 18.8    |
| 2                                             | Net Profit = 0 | 0    | 0.0     |

Figure 9: Net Profit Distribution

As shown in Figure 9, out of 133 operating days, 108 days (81.2%) recorded positive Net profit, while 25 days (18.8%) resulted in losses. No day recorded exactly zero profit.

At first glance, this suggests that the café is profitable on most days. However, this statistic alone does not guarantee stability. Even though the majority of days are profitable, the presence of nearly 19% loss days indicates recurring financial stress.

More importantly, as shown in Figure 6 and Figure 14, loss days are not minor dips but often involve significant negative values. This means that a relatively small proportion of days can materially affect monthly performance.

This distribution supports an important insight: the café does not suffer from consistent underperformance, but from periodic instability driven by specific high-cost days.

### 3.2.4 Volatility Assessment Using Coefficient of Variation

| === Volatility (Coefficient of Variation) === |          |           |           |                    |
|-----------------------------------------------|----------|-----------|-----------|--------------------|
|                                               | Variable | Mean      | StdDev    | Coeff_of_Variation |
| 0                                             | Sales    | 1490.7293 | 955.3738  | 0.6409             |
| 1                                             | Profit   | 39.5940   | 3617.9870 | 91.3772            |

Figure 10: Coefficient of Variation

As shown in Figure 10, the coefficient of variation (CV) was calculated for both Sales and Profit to measure relative instability.

Sales show a coefficient of variation of 0.64, indicating moderate variability relative to the average sales level. While daily sales fluctuate, the variation remains within a proportional range.

However, Profit shows an extremely high coefficient of variation of **91.38**. This means that profit fluctuates far more aggressively relative to its average value. The reason for this extreme value is that the average profit is very low compared to its standard deviation, indicating large swings between positive and negative outcomes.

This difference between sales volatility and profit volatility is critical. It suggests that instability is not primarily demand-driven. Instead, profit is highly sensitive to cost behaviour. Small changes in expenses, especially spikes, cause disproportionate impacts on earnings.

The coefficient of variation therefore confirms that the café's financial instability is structural at the profit level, even if revenue itself does not fluctuate at an extreme scale.

### 3.2.5 Monthly Net Profit Distribution

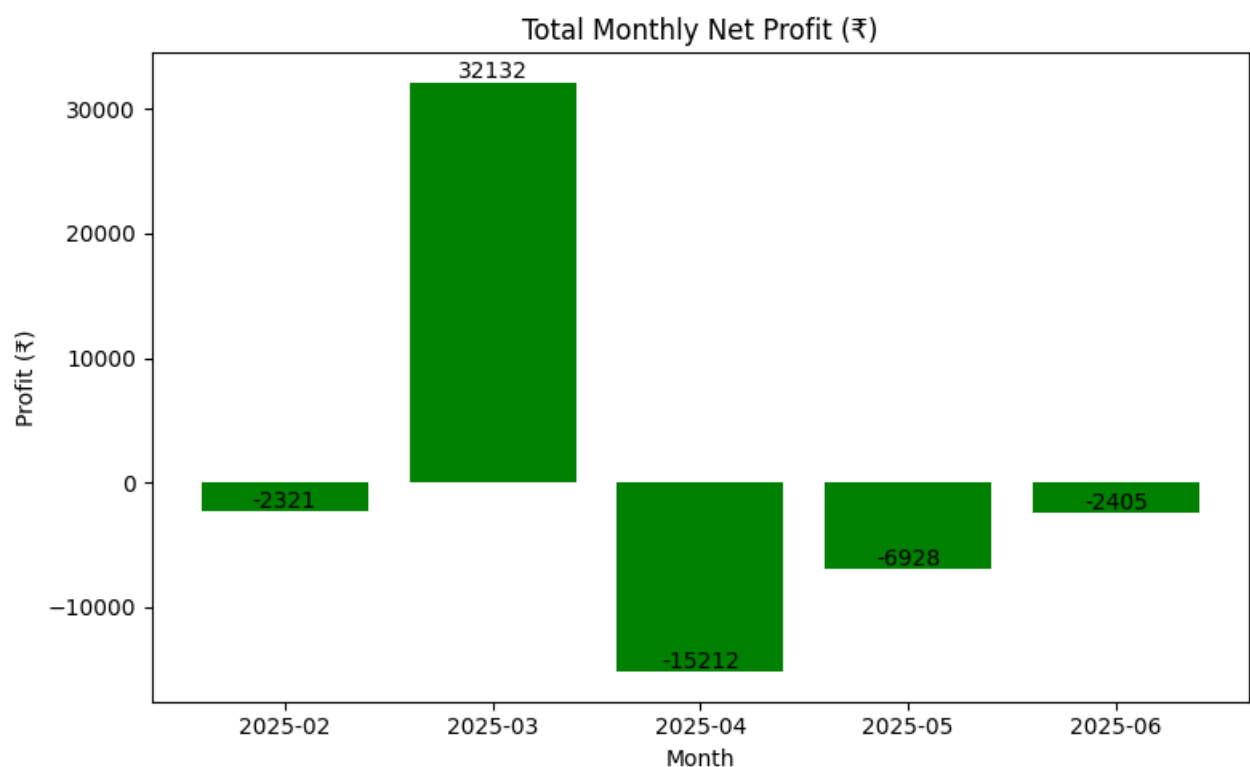


Figure 11: Month-wise Net Profit Distribution

As Figure 11 shows, if we aggregate profit at a monthly level, the instability becomes even more evident.

March records a strong positive profit of ₹32,132. However, all other months show negative totals, with April being particularly severe at –₹15,212. Even February, May and June record overall losses.

This pattern is important because it shows that monthly outcomes are highly sensitive to a few extreme days. The café does not generate steady month-to-month earnings. Instead, performance is uneven, with one strong month offset by multiple weak ones.

This explains the owner's difficulty in estimating monthly income. Even if many individual days are profitable, irregular expense spikes and loss days significantly distort month-end results.

The monthly aggregation confirms that financial instability is not limited to daily fluctuations but extends to longer-term performance as well.

### 3.2.6 Net Profit Distribution Analysis

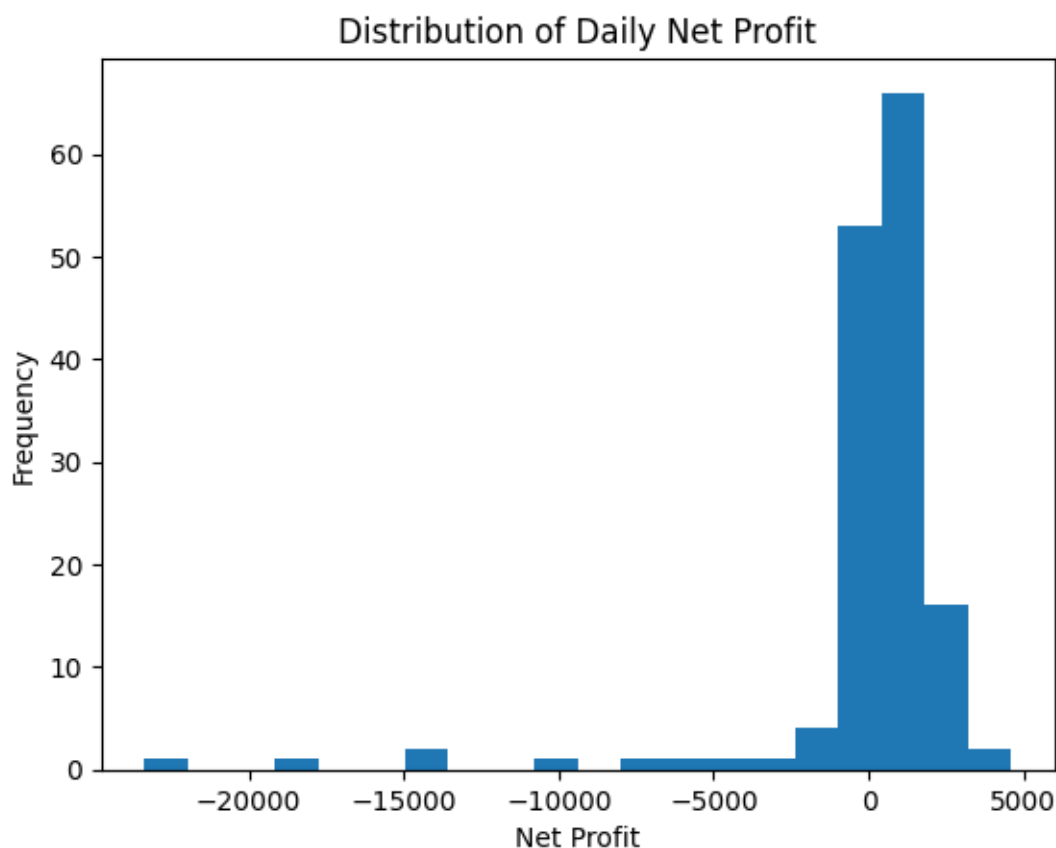


Figure 12: Distribution of Daily Net Profit

As shown in Figure 12, the distribution of daily profits shows an imbalance of financial performance. Most values are concentrated around small positive values which indicates that the café operates at a decent profit on most days.

However, the distribution has a long negative tail, where a few observations extend into very large losses. These extreme values are significantly larger in magnitude than typical daily profits.

This confirms that the instability in overall performance is not due to frequent losses, but due to the magnitude of a few extreme negative days. These rare but large losses outweigh the gains from regular operations.

### 3.2.7 Weekly Net Profit Stability

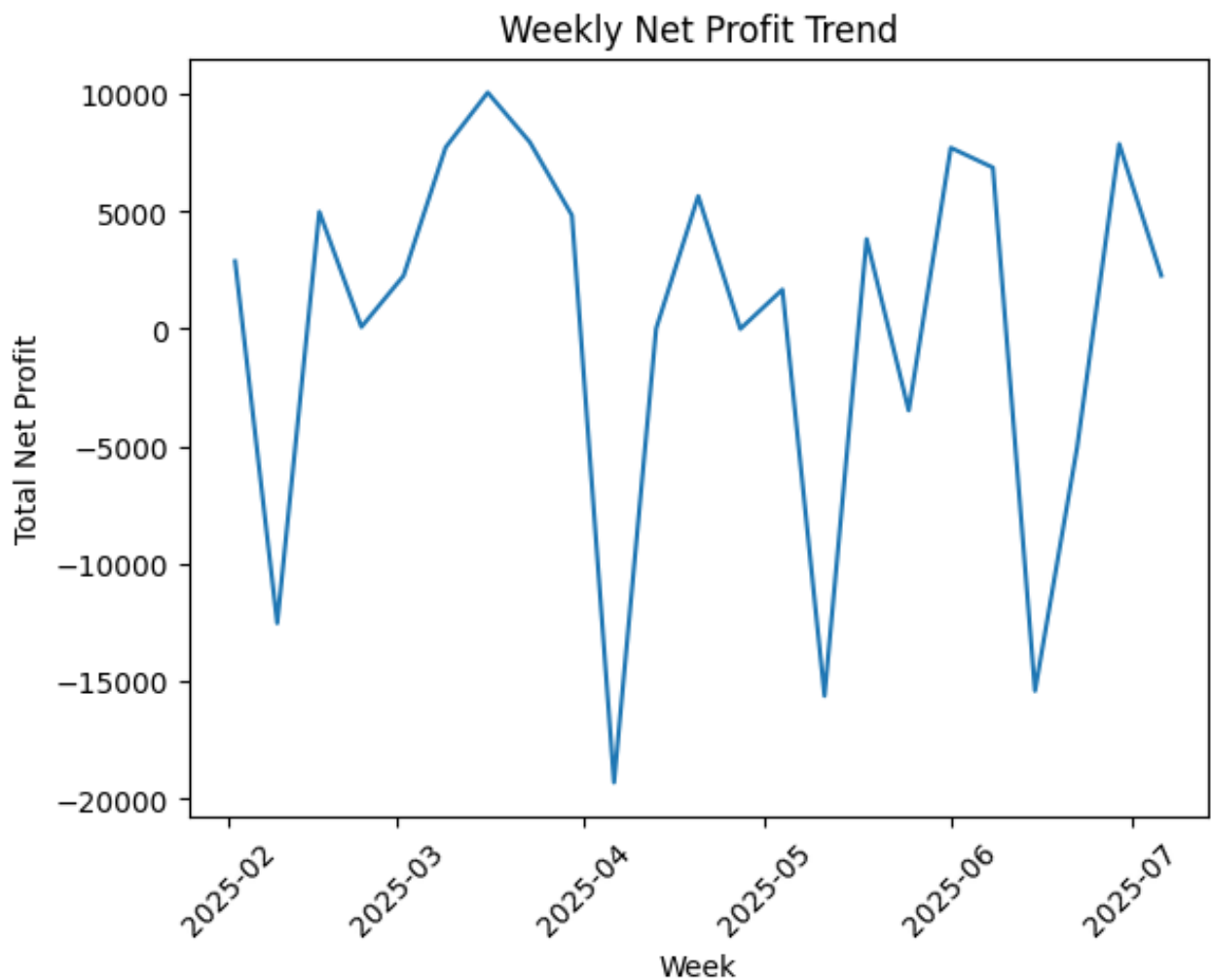


Figure 13: Weekly Net Profit Trend

As shown in Figure 13, when net profit is aggregated at a weekly level, short-term fluctuations are reduced and broader patterns become visible.

Despite this aggregation, the weekly trend still has multiple negative total profits. This indicates that the instability isn't just at a daily level but visible at longer intervals.

The weekly performance suggests that the volatility is structural rather than random due to multiple negative spikes. The café isn't maintaining stable earnings at a weekly level.

### 3.2.8 Loss Magnitude Analysis

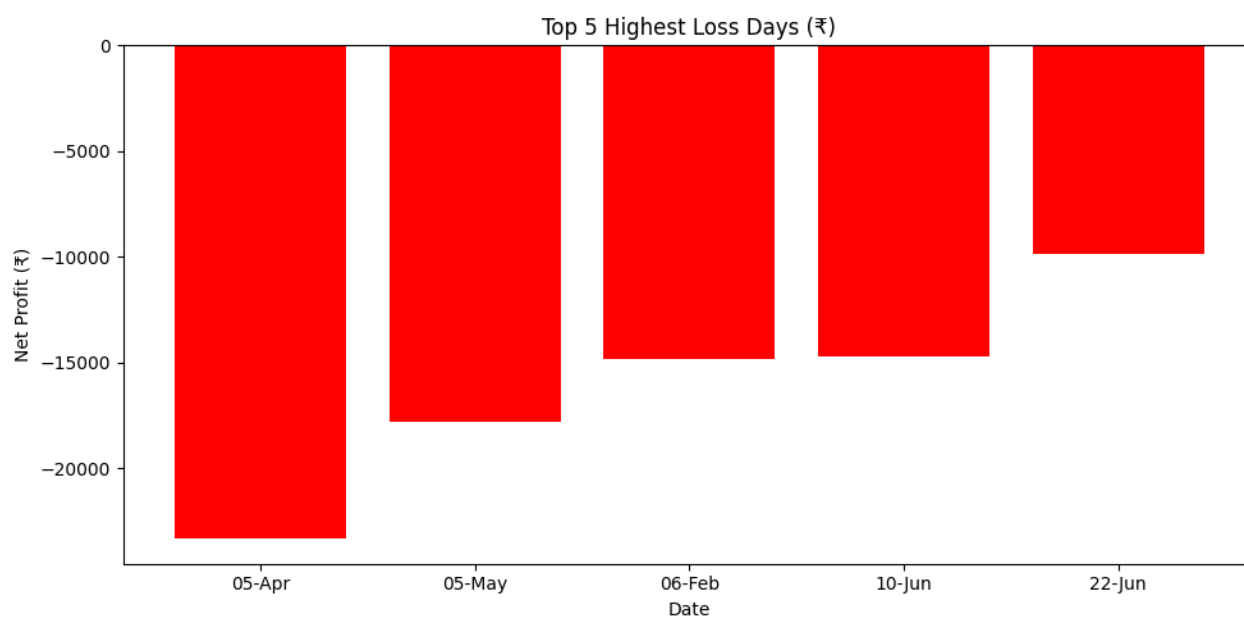


Figure 14: Top 5 Highest Loss Days

As shown in Figure 14, the top loss days show extremely large negative values compared to typical daily profits.

These losses are major financial shocks. Each of these days individually has a strong impact on overall monthly performance.

This reinforces the earlier finding that controlling extreme loss events is more important than improving average daily profit.

## **4 Interpretation of Results and Recommendations**

To summarize everything, the café's problem is not about people not coming to their shop. They have quite a few sales every day, especially for a newly opened café. The problem is that whenever a fixed cost like worker salary, rent, bulk delivery etc. hit they take away most of the earning in a week. My recommendations below hope to help the owner tackle this issue.

Even after aggregating profit weekly, earnings remain inconsistent. This shows that instability is not limited to individual days. Profit distribution shows that while most days generate small positive profits, a few extreme loss days have a disproportionately large impact. Break-even analysis further highlighted that many days operate with a very small margin of safety, making the business highly sensitive to cost increases.

These observations tell us that the café's performance is driven more by irregular expense behaviour than by demand fluctuations.

Based on these findings, the following recommendations are made.

### **4.1 Redesign Salary and Fixed Cost Timing**

As shown in Figures 6, 11 and 14, the scatter plot and monthly loss pattern suggest that large expense spikes are concentrated on specific days. These are likely bulk payments such as salaries or major procurement.

Instead of recording large lump-sum expenses on a single day, fixed costs should be:

- Spread weekly instead of monthly where possible
- Allocated proportionally across accounting days

This will not change total cost, but it will reduce extreme daily loss spikes, lowering profit volatility and stabilizing monthly outcomes.

### **4.2 Align Bulk Procurement with High-Demand Windows**

As shown in Figure 8, Friday shows the highest average net profit despite not being the top sales day. Figure 4 confirms that Saturday leads in average sales but, as Figure 8 shows, this does not translate into profitability.

Bulk inventory purchases should be scheduled closer to historically strong demand days. This ensures that high procurement expense is supported by higher revenue potential.

The data shows that loss days are not always low-sales days. Therefore, procurement timing should be demand-aligned rather than calendar-based.

### **4.3 Introduce a Daily Expense Ceiling Based on Revenue Bands**

As shown in Figures 2 and 10, profit becomes unstable when expense exceeds a large proportion of sales.

A practical rule can be implemented:

- If daily sales fall below a certain threshold (for example, below historical median sales), discretionary spending should be reduced.
- If sales exceed break-even early in the day, additional procurement may be safer.

This creates adaptive cost behaviour rather than fixed spending patterns.

### **4.4 Focus on Reducing Loss Magnitude, Not Just Loss Frequency**

As shown in Figures 9, 12 and 14, only 18.8% of days are loss-making, but these losses are significantly larger than typical daily profits.

The priority should therefore be:

- Preventing large negative spikes
- Identifying the top 5 highest-loss days each month
- Investigating their expense composition

Reducing the magnitude of extreme losses will improve monthly stability more effectively than trying to eliminate every minor negative day.

### **4.5 Use Profit Volatility as a Performance Indicator**

As shown in Figure 10, the coefficient of variation for net profit (91.38) is significantly higher than for sales (0.64).

Instead of focusing only on total monthly profit, the owner should track:

- Standard deviation of daily profit
- Maximum single-day loss
- Number of days exceeding historical expense average

The goal should be to reduce volatility over time, not just increase revenue.

#### **4.6 Re-evaluate Weekend Cost Behaviour**

As shown in Figures 4 and 8, although Saturday has the highest average sales, it records negative average net profit. This indicates cost inefficiency on a high-demand day.

Operational practices on Saturdays should be reviewed specifically:

- Overstaffing
- Excess Inventory
- Wastage

Improving efficiency on high-demand days can increase profitability without increasing sales.

#### **4.7 Implement Consistent Item-Level Sales Recording**

The current sales data is recorded only as a daily total, making it impossible to identify which items contribute most to revenue or which products have the highest margins. On some days, item-level details were available, but this was inconsistent across the observation period. The business should begin maintaining a simple daily tally of items sold, recording item name, quantity, and price per unit. Even a basic handwritten log would be sufficient. This would enable meaningful sales composition analysis in future cycles and allow the owner to make informed decisions about menu pricing and product focus.

#### **4.8 Implement Consistent Per-Category Expense Recording**

While expense breakdown data by category (raw materials, delivery, salary, rent, Google Ads) was available for some days, it was missing on the majority of operating days. A reliable daily record of expense components would allow the owner to track which cost categories are growing over time and identify specific areas for reduction. A simple template recording each expense category at the end of each day would be sufficient. This would significantly improve the quality of future financial analysis and enable more targeted cost control decisions.

## **5 Additional**

This section includes some points that I thought were worth mentioning over and above the main analysis above.

### **5.1 Limitations and Scope for Improvement**

This analysis is based on daily aggregated data of sales and expenses, which limits the level of detail that can be explored. Since item-level or category-wise data is not available consistently, it is not possible to identify which products contribute most to revenue or which cost components drive inefficiencies.

Although expense data has been collected on a per-category basis (raw materials, delivery, worker salary, rent, and Google Ads), this breakdown is not consistently available across all operating days. More than 50% of days have partial or missing per-category entries. On most normal operating days, the total expense figure is recorded but its internal composition is not. Including per-category expense analysis under these conditions would risk presenting factually incorrect breakdowns for a majority of observations, which would undermine the integrity of the findings. The aggregate expense figure has therefore been retained as the primary variable, with category level data used only where reliably recorded.

Item-level sales data was also inconsistently available across the observation period. While some days had detailed records of individual items sold, this information was absent on the majority of days, making it impossible to conduct reliable item-wise analysis. Including partial item-level data would risk presenting incomplete and potentially misleading findings for a significant proportion of operating days, which would undermine the integrity of the analysis. Daily sales totals have therefore been retained as the primary sales variable.

Additionally, external factors such as seasonality, promotions, or changes in customer behaviour are not explicitly captured in the dataset. These factors may influence sales patterns but cannot be isolated in the current analysis.

Future analysis can be improved by collecting more granular data, such as item-wise sales, cost categories, and time-of-day transactions. This would allow more precise identification of inefficiencies and enable more targeted decision-making.

## 5.2 Early-Stage Behavior

The data used in this analysis was collected when the café was in its early days of operation. At this stage, businesses typically experience irregular cost patterns due to setup activities, supplier adjustments and experimental inventory levels.

This is also evident in the dataset in the form of high expense days (days where we see expenses in the tens of thousands), especially in the early days. These expenses include one-time costs like equipment purchase, brokerage for property, seating equipment, cooking equipment, supplier agreements and initial stock buildup. This detail was also told to me by the owner in my interaction with him while I was trying to understand their problem. They made it clear that these high expenses were a normal occurrence for them as it's not their first venture and they expect costs to be high initially.

Although, even after we account for the early-stage behavior, the profit level variability remains significant. The persistence of large fluctuations beyond isolated setup-related costs suggests that instability is not solely due to the café being new, but also due to the way expenses are timed and managed.

This distinction is important because it indicates that while some level of instability is expected in the beginning, structured cost control practices are still necessary to ensure consistent performance as the business matures.

## 5.3 Additional Links

- [Working Dataset](#)
- [Colab Notebook](#)